



Instructor: Date:

Learner: Regional Branch:

Program: Center:

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Let's Start!

Optimizing the Programming Task

```
<!DOCTYPE html>
<html>
  <head>
    .
  </head>
  <body>
    .
    .
  </body>
</html>
```

Learning Outcome:

Talk about principles, concepts, characteristics of general-purpose computer programming language in spoken and written situations.

Ask your partner:

- How can you save time when writing a code?
- How can you develop different classes with similar characteristics?

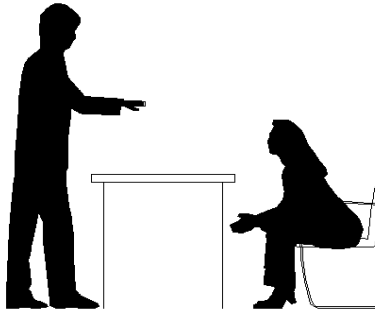
Can Do Statements:

- I can talk about inheritance in C#.
- I can discuss information about base and derived classes in C#.
- I can say how to implement multiple inheritance in C#.



Let's Explore!

You might want to use inheritance in C# language.



Instructions: Read the text below.

Ruth: Lovely to see you, Thomas.
Thomas: Nice to see you too, Ruth.
Ruth: Can I offer you something; coffee, tea?
Thomas: Tea, please.
Ruth: Would you rather have black or white tea?
Thomas: Black tea, thank you.
Ruth: Okay. Thomas, I called you because I need your help with a programming task.
Thomas: Sure. Would you rather program with C# or Java?
Ruth: C#. I already have some classes, and the new ones would share certain features on their behavior.
Thomas: You might want to use inheritance, then. It is one of the pillars in object-oriented programming.
Ruth: Inheritance? I have barely heard about it. I only know there exist single and multiple inheritance.
Thomas: Well, Ruth. Inheritance is a way in which you can copy and modify the behavior of an existing class and reuse it in a new class; this way you can save time and keep the code functionality. When working with inheritance, we have two classes: base and derived. By using the variables and members contained in the base class, the new or "derived class" will inherit the behavior features of the base class. Now, talking about single and multiple inheritance, the first one takes characteristics from a single base class, and the second one takes those characteristics from several bases.
Ruth: Awesome. I will use multiple inheritance, then.
Thomas: Sorry, Ruth. C# does not support multiple inheritance.
Ruth: I thought it was possible.
Thomas: Well, it's possible, but you need the implementation of interfaces to declare a multiple inheritance. When you declare it, the derived class can inherit characteristics and features from different base classes.
Ruth: Thank you very much, Thomas. You're awesome. I will read more about it to start with the programming task.
Thomas: You're welcome, Ruth. Would you rather read from a book or the internet?
Ruth: I would prefer a book; I don't really like reading from the screen.
Thomas: Okay. I can lend you a book. You had better start reading from page 66; there is where the topic starts.

**Let's Practice!**

Ruth receives good advice.

Instructions: Type in the missing words. Use might want to learn, ought to, might want, had better use. Use capital letters where necessary.

1. You _____ inheritance to save time.
2. You _____ to know how to use inheritance properties in programming.
3. I think Thomas _____ help Ruth if she doesn't understand the topic well.
4. Ruth _____ to work with interfaces.

**Let's Practice!**

What do you prefer?

Instructions: Use the words from the word bank to complete each sentence.

Word Bank

Rather read / would / rather use / you rather

1. Which language program would Ruth _____, C# or Java?
2. What inheritance _____ Ruth rather use, single or multiple?
3. Would you _____ from a book or a screen?
4. Would _____ use existing code or start everything from scratch?



Extension Activity

Applying inheritance in C# programming language.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Research what codes for inheritance implementation can be used to transfer attributes from a base class to a derived class in C# language.
- Write a set of instructions in a base class that lead to a concrete result.
- Use the inheritance codes already investigated and add more functionalities to the derived class you are creating.
- Present the result in a piece of paper. Explain what you did if necessary.

Self-Assessment

Yes

No

- I look for information on inheritance in C# language programming.
- I list characteristics of inheritance in C#.
- I recognize how to use interfaces to work with inheritance in C#.

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Target Words

- **reuse** (verb): to use again
e.g. Inheritance is a way in which you can copy and modify the behavior of an existing class and reuse it in a new class.
- **code functionality** (noun): a series of instructions intended to produce a result
e.g. This way you can save time and keep the code functionality.
- **support** (verb): in computing, to allow to do something; to be compatible
e.g. Sorry, Ruth. C# does not support multiple inheritance.
- **barely** (adverb): in a scarce amount
e.g. I have barely heard about it. I only know there exist single and multiple inheritance.
- **inherit** (verb): to receive elements or features from something or someone who came to existence before the receiver
e.g. By using the variables and members contained in the base class, the new or "derived class" will inherit the behavior features of the base class.



Learning Points

Would Rather for What Is Preferable - Interrogative Form.

You use **Would Rather** to say you prefer one thing to another.

We can ask people to decide between -given options by using **would rather** in the interrogative form.

Question structures:

Would + subject + rather + base form of the verb + option 1 + or + option 2?

- *Would you rather have tea or coffee? Would you rather use C# language or Java in your programming assignment?*

Wh + would + subject + rather + base form of the verb + option 1 + or + option 2?

- *Which property would you rather use, single or multiple inheritance?*
- *Where would you rather work next year, in an IT department of a local company or in another area with a multinational company?*

Examples

- *Would you rather have black or white tea?*
- *Would you rather program with C# or Java?*
- *Would you rather read from a book or the internet?*

Understanding the Concept of Multiple Inheritance

In programming languages, you will find the concept of "inheritance."

When we manage programming languages, we can talk about two types of inheritance: single and multiple. **Multiple inheritance** appears when a derived object or class adopts characteristics and features from more than one base object or class. In contrast, single inheritance allows a derived object or class to inherit from only one base object or class.

When working in C#, we are not allowed to work directly with **multiple inheritance**; instead, we can resort to **interfaces** to do this task.

